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# Functional Equations

*An Original Olympiad Practice Set*

**compiled & written by Indrajeet Yadav**

**Original content.** These 8 problems were written fresh for this set — they are **not** copied from Titu Andreescu's books or any competition. An original eight-problem ladder through real-valued functional equations: Cauchy additivity, the swap and reciprocal-cycle substitution tricks, exploiting injectivity and surjectivity, and pinning down functions via fixed points. Each problem turns on a single decisive idea rather than computation, ordered from warm-up to hard olympiad. Every closed-form answer was checked symbolically/numerically. Free to share for study.

# Problems

**1.** DIFFICULTY 1/5 · CAUCHY EQUATION, INTEGER SCALING

A function $f : \mathbb{R} \to \mathbb{R}$ satisfies $f(x+y) = f(x) + f(y)$ for all real $x, y$, and $f(3) = 12$. Find $f(10)$, and show this value is forced even without assuming $f$ is continuous.

**2.** DIFFICULTY 2/5 · SWAP SUBSTITUTION, LINEAR SYSTEM, REFLECTION X->2-X

Find all functions $f : \mathbb{R} \to \mathbb{R}$ satisfying

$$f(x) - 2f(2-x) = x^2 \qquad \text{for all } x \in \mathbb{R},$$

and compute $f(0)$.

**3.** DIFFICULTY 3/5 · MODIFIED CAUCHY, SUBTRACT A PARTICULAR SOLUTION, INTEGER DETERMINACY

A function $f : \mathbb{R} \to \mathbb{R}$ satisfies

$$f(x+y) = f(x) + f(y) + xy(x+y) \qquad \text{for all } x, y \in \mathbb{R},$$

and $f(1) = 1$. Find $f(3)$.

**4.** DIFFICULTY 3/5 · FUNCTIONAL EQUATION AS RECURRENCE, CHARACTERISTIC EQUATION, FIXED POINTS

Find all functions $f : \mathbb{R} \to \mathbb{R}$ of the form $f(x) = cx$ (a real constant $c$) for which

$$f\big(f(x)\big) + f(x) = 6x \qquad \text{for all } x \in \mathbb{R}.$$

For each such $f$, determine all real fixed points (solutions of $f(x) = x$).

**5.** DIFFICULTY 4/5 · RECIPROCAL-CYCLE SUBSTITUTION, ORDER-3 ORBIT, LINEAR SYSTEM

Find all functions $f$ defined on $\mathbb{R} \setminus \{0, 1\}$ satisfying

$$f(x) + f\left(\frac{1}{1-x}\right) = x \qquad \text{for all } x \neq 0, 1,$$

and evaluate $f(2)$.

**6.** DIFFICULTY 4/5 · D'ALEMBERT-TYPE EQUATION, EVENNESS, EVALUATION AT 0, CONTINUOUS SOLUTIONS

A continuous function $f : \mathbb{R} \to \mathbb{R}$, not identically zero, satisfies

$$f(x)f(y) = f(x+y) + f(x-y) \qquad \text{for all } x, y \in \mathbb{R}.$$

Prove that $f$ is even and that $f(0) = 2$, and describe all such $f$.

**7.** DIFFICULTY 4/5 · ADDITIVITY, MULTIPLICATIVITY VIA SQUARES, MONOTONICITY FROM NONNEGATIVITY

Find all functions $f : \mathbb{R} \to \mathbb{R}$ satisfying both

$$f(x+y) = f(x) + f(y) \quad \text{and} \quad f\left(x^2\right) = f(x)^2 \qquad \text{for all } x, y \in \mathbb{R}.$$

(No continuity is assumed.)

**8.** DIFFICULTY 5/5 · SURJECTIVITY, INJECTIVITY, FIXED-POINT CHASE, CAUCHY ON A GENERATED SET

Find all functions $f : \mathbb{R} \to \mathbb{R}$ satisfying

$$f\left(x^2 + f(y)\right) = y + \left(f(x)\right)^2 \qquad \text{for all } x, y \in \mathbb{R}.$$

# Solutions

**1. Answer:** $f(10) = 40$

Set $x = y = 0$: $f(0) = 2f(0)$, so $f(0) = 0$. For any real $a$ and positive integer $n$, repeated use of additivity gives $f(na) = nf(a)$ by induction: $f((n+1)a) = f(na) + f(a) = nf(a) + f(a) = (n+1)f(a)$. Apply with $a = 1,\ n = 3$: $12 = f(3) = f(1+1+1) = 3f(1) \Rightarrow f(1) = 4$. Then $f(10) = f(10 \cdot 1) = 10f(1) = 40$. No continuity is needed: the value at any integer multiple of a base point is determined purely by additivity, and 10 is an integer multiple of 1. Hence $f(10) = 40$ is forced. (The general additive solution can be wildly discontinuous, but it is still pinned down at every rational, so all integer values are determined.)

**Second method.** Directly: $f(10) = f(3+3+3+1) = 3f(3) + f(1)$. Since $f(3) = 3f(1) = 12$ gives $f(1) = 4$, we get $f(10) = 3 \cdot 12 + 4 = 40$. Equivalently $f(10) = f(3) + f(3) + f(3) + f(1) = 12 + 12 + 12 + 4 = 40$.

**2. Answer:** $f(x) = -x^2 + \dfrac{8x}{3} - \dfrac{8}{3}$**, so** $f(0) = -\dfrac{8}{3}$

The substitution $x \mapsto 2 - x$ is an involution, since $2 - (2 - x) = x$; applying it twice returns to the start. Write the given equation and its image under $x \mapsto 2 - x$:

$$f(x) - 2f(2-x) = x^2 \tag{A}$$

$$f(2-x) - 2f(x) = (2-x)^2. \tag{B}$$

Treat $f(x)$ and $f(2-x)$ as two unknowns. From (A), $f(x) = x^2 + 2f(2-x)$. Substitute into (B):

$$f(2-x) - 2\big(x^2 + 2f(2-x)\big) = (2-x)^2 \ \Rightarrow\ -3f(2-x) = (2-x)^2 + 2x^2,$$

so $f(2-x) = -\dfrac{(2-x)^2 + 2x^2}{3}$. Then

$$f(x) = x^2 + 2f(2-x) = x^2 - \tfrac{2}{3}\big((2-x)^2 + 2x^2\big).$$

Expanding, $(2-x)^2 + 2x^2 = 3x^2 - 4x + 4$, so $f(x) = x^2 - \frac{2}{3}(3x^2 - 4x + 4) = -x^2 + \dfrac{8x}{3} - \dfrac{8}{3}$. A quick check confirms (A). Hence $f(0) = -\dfrac{8}{3}$. The solution is unique because the $2 \times 2$ linear system has determinant $(1)(1) - (-2)(-2) = -3 \neq 0$.

**Second method.** Ansatz: since the right side is quadratic and the transformation is affine, seek $f(x) = ax^2 + bx + c$. Then $f(x) - 2f(2-x) = ax^2 + bx + c - 2\big(a(2-x)^2 + b(2-x) + c\big)$. Matching the $x^2$ coefficient gives $a - 2a = 1 \Rightarrow a = -1$; matching the $x$- and constant coefficients then yields $b = \frac{8}{3},\ c = -\frac{8}{3}$, the same function.

**3. Answer:** $f(3) = 11$

The cubic-looking term $xy(x+y)$ is exactly the obstruction produced by a cubic. Note $(x+y)^3 = x^3 + y^3 + 3xy(x+y)$, so $p(x) = \frac{1}{3}x^3$ satisfies

$$p(x+y) - p(x) - p(y) = \tfrac{1}{3}\big((x+y)^3 - x^3 - y^3\big) = xy(x+y),$$

i.e. $p$ is a particular solution of the inhomogeneous equation. Set $g(x) = f(x) - \frac{1}{3}x^3$. Subtracting the two relations,

$$g(x+y) = \big(f(x) + f(y) + xy(x+y)\big) - \big(p(x) + p(y) + xy(x+y)\big) = g(x) + g(y),$$

so $g$ is additive (Cauchy). Hence $g(3) = 3g(1)$ (additivity forces this at every integer, no continuity needed). Now $g(1) = f(1) - \frac{1}{3} = 1 - \frac{1}{3} = \frac{2}{3}$, so $g(3) = 2$. Therefore

$$f(3) = g(3) + \tfrac{1}{3} \cdot 27 = 2 + 9 = 11.$$

**Second method.** Build up directly. Put $y = 1$: $f(x+1) = f(x) + 1 + x^2 + x$. From $f(1) = 1$: with $x = 1$, $f(2) = f(1) + 1 + 1 + 1 = 4$; with $x = 2$, $f(3) = f(2) + 1 + 4 + 2 = 4 + 7 = 11$. Same answer, $f(3) = 11$.

**4. Answer: $f(x) = 2x$ (fixed point only $x = 0$) and $f(x) = -3x$ (fixed point only $x = 0$)**

With $f(x) = cx$ we have $f(f(x)) = c(cx) = c^2x$, so the equation becomes $c^2x + cx = 6x$ for all $x$, i.e.

$$c^2 + c - 6 = 0 \implies (c-2)(c+3) = 0 \implies c = 2 \text{ or } c = -3.$$

Thus the linear solutions are exactly $f(x) = 2x$ and $f(x) = -3x$. The characteristic equation $c^2 + c - 6 = 0$ is precisely the relation you would get by treating the iterate $f \circ f$ like a second-order recurrence on the slope. Fixed points: solve $cx = x$, i.e. $(c-1)x = 0$. Since $c = 2$ gives $c - 1 = 1 \neq 0$ and $c = -3$ gives $c - 1 = -4 \neq 0$, in both cases the only solution is $x = 0$. So each map has the unique fixed point $x = 0$. (Insight: any such $f$ must fix 0: putting $x = 0$ gives $f(f(0)) + f(0) = 0$, automatic for a linear map; the slope condition then isolates the two admissible rates.)

**Second method.** View the equation as forcing the operator identity $f^2 + f - 6\,\mathrm{id} = 0$ on the slope, where $f^2 = f \circ f$. Factor: $(f - 2\,\mathrm{id})(f + 3\,\mathrm{id}) = 0$. For a scalar multiplier $c$ this annihilator means $c = 2$ or $c = -3$, matching the roots above; the fixed-point set of $x \mapsto cx$ with $c \neq 1$ is $\{0\}$.

**5. Answer:** $f(x) = \dfrac{x^3 - x + 1}{2x(x-1)}$**, and** $f(2) = \dfrac{7}{4}$

The key observation: the substitution $T(x) = \dfrac{1}{1-x}$ has order three. Compute $T(T(x)) = \dfrac{1}{1 - \frac{1}{1-x}} = \dfrac{x-1}{x}$, and $T(T(T(x))) = x$. So the three points $x$, $T(x) = \dfrac{1}{1-x}$, $T^2(x) = \dfrac{x-1}{x}$ form a 3-cycle, all lying in the domain. Apply the equation at these three arguments:

$$f(x) + f\left(\tfrac{1}{1-x}\right) = x, \tag{1}$$

$$f\left(\tfrac{1}{1-x}\right) + f\left(\tfrac{x-1}{x}\right) = \tfrac{1}{1-x}, \tag{2}$$

$$f\left(\tfrac{x-1}{x}\right) + f(x) = \tfrac{x-1}{x}. \tag{3}$$

Form $(1) - (2) + (3)$: the middle terms cancel and $2f(x) = x - \dfrac{1}{1-x} + \dfrac{x-1}{x}$. Hence

$$f(x) = \frac{1}{2}\left(x - \frac{1}{1-x} + \frac{x-1}{x}\right) = \frac{x^3 - x + 1}{2x(x-1)}.$$

This satisfies (1) by construction, and uniqueness follows since the $3 \times 3$ cyclic system is nonsingular. Finally $f(2) = \dfrac{8 - 2 + 1}{2 \cdot 2 \cdot 1} = \dfrac{7}{4}$.

**Second method.** Denote $a = f(x)$, $b = f(T(x))$, $d = f(T^2(x))$, with right-hand sides $x$, $\frac{1}{1-x}$, $\frac{x-1}{x}$ summing to $S$. Adding all three equations gives $2(a + b + d) = S$, so $a + b + d = \frac{S}{2}$; subtracting equation (2) (which is $b + d = \frac{1}{1-x}$) isolates $a = f(x) = \frac{S}{2} - \frac{1}{1-x}$. At $x = 2$, $S = 2 - 1 + \frac{1}{2} = \frac{3}{2}$, so $f(2) = \frac{3}{4} - (-1) = \frac{7}{4}$.

**6. Answer:** $f(0) = 2$**; the continuous nonzero solutions are** $f(x) = 2\cos(ax)$ **and** $f(x) = 2\cosh(ax)$ **for some real** $a \geq 0$ **(with** $a = 0$ **giving** $f \equiv 2$**).**

Step 1 (value at 0). Put $x = y = 0$: $f(0)^2 = f(0) + f(0) = 2f(0)$, so $f(0) \in \{0, 2\}$. If $f(0) = 0$, set $y = 0$ in the original: $f(x)f(0) = f(x) + f(x)$, i.e. $0 = 2f(x)$, forcing $f \equiv 0$, excluded. Hence $f(0) = 2$. Step 2 (evenness). Put $x = 0$: $f(0)f(y) = f(y) + f(-y)$, i.e. $2f(y) = f(y) + f(-y)$, so $f(-y) = f(y)$; $f$ is even. Step 3 (classification). The relation $f(x)f(y) = f(x+y) + f(x-y)$ is d'Alembert's equation (normalized so $f(0) = 2$). Write $g = f/2$; then $g(x)g(y) = \frac{1}{2}\big(g(x+y) + g(x-y)\big)$, the standard cosine equation, whose continuous solutions are $g(x) = \cos(ax)$ or $g(x) = \cosh(ax)$ for some $a \geq 0$. Therefore $f(x) = 2\cos(ax)$ or $f(x) = 2\cosh(ax)$; the degenerate $a = 0$ case is $f \equiv 2$. Verification: for $f = 2\cos(ax)$, $f(x)f(y) = 4\cos ax \cos ay = 2\big(\cos a(x+y) + \cos a(x-y)\big) = f(x+y) + f(x-y)$; the cosh case uses $\cosh u \cosh v = \frac{1}{2}(\cosh(u+v) + \cosh(u-v))$. Thus all continuous nonzero solutions are exactly these.

**Second method.** Without quoting d'Alembert: evenness plus the substitution $y = x$ gives $f(x)^2 = f(2x) + f(0) = f(2x) + 2$, a doubling formula $f(2x) = f(x)^2 - 2$ — exactly the identity satisfied by $2\cos$ ($2\cos 2\theta = (2\cos\theta)^2 - 2$) and $2\cosh$. Continuity then upgrades determinacy to the closed forms; in all cases $f(0) = 2$ and $f$ is even, as required.

**7. Answer: $f \equiv 0$ and $f(x) = x$**

Additivity gives $f(0) = 0$ and $f(qx) = qf(x)$ for all rationals $q$; in particular $f$ restricted to $\mathbb{Q}$ is $q \mapsto cq$ with $c = f(1)$. Step 1 (sign on nonnegatives). For $t \geq 0$ write $t = x^2$; then $f(t) = f(x^2) = f(x)^2 \geq 0$. So $f$ is nonnegative on $[0, \infty)$. Step 2 (monotonicity). If $a \leq b$, then $b - a \geq 0$, so $f(b) - f(a) = f(b - a) \geq 0$ by additivity and Step 1. Thus $f$ is nondecreasing. An additive function that is monotone is linear: combined with $f(q) = cq$ on rationals, monotonicity forces $f(x) = cx$ for all real $x$ (squeeze $x$ between rationals from both sides). Step 3 (pin the slope). The square condition gives $f(1) = f(1)^2$, i.e. $c = c^2$, so $c \in \{0, 1\}$. Therefore $f \equiv 0$ or $f(x) = x$. Both clearly satisfy the two equations. The decisive insight is that the square condition is a hidden positivity/monotonicity hypothesis — it tames the otherwise pathological additive solutions without invoking continuity.

**Second method.** Alternative for Step 3 via multiplicativity: additivity and $f(x^2) = f(x)^2$ together yield $f(xy) = f(x)f(y)$. Indeed $f((x+y)^2) = f(x+y)^2$ expands by additivity to $f(x^2) + 2f(xy) + f(y^2) = f(x)^2 + 2f(x)f(y) + f(y)^2$; cancelling $f(x^2) = f(x)^2$ and $f(y^2) = f(y)^2$ gives $f(xy) = f(x)f(y)$. A function both additive and multiplicative on $\mathbb{R}$ is either 0 or the identity (multiplicativity gives $c = c^2$ and preserves order via squares, recovering $f = 0$ or $f = \mathrm{id}$).

**8. Answer:** $f(x) = x$

Step 1 (bijectivity). Fix $x = 0$: $f(f(y)) = y + f(0)^2$. The right side is a bijection of $y$, so $f \circ f$ is a bijection; hence $f$ is injective (if $f(y_1) = f(y_2)$ then $f(f(y_1)) = f(f(y_2)) \Rightarrow y_1 = y_2$) and surjective (since $f \circ f$ is onto, so is $f$). Surjectivity also follows directly: fixing any $x$, the right side $y + f(x)^2$ ranges over all reals. Step 2 (locate a zero and a key relation). By surjectivity choose $c$ with $f(c) = 0$. Put $x = c$:

$$f\big(c^2 + f(y)\big) = y + f(c)^2 = y. \tag{$*$}$$

Put $y = c$ in the original: $f(x^2 + f(c)) = c + f(x)^2$, i.e.

$$f(x^2) = c + f(x)^2. \tag{$\dagger$}$$

Setting $x = 0$ in (\dagger) and writing $d = f(0)$ gives $d = c + d^2$. Step 3 (show $c = 0$). Apply $f$ to (\ast) using Step 1's relation $f(f(\cdot)) = (\cdot) + d^2$. From (\ast) with $y = c$: $f(c^2) = c$. From (\dagger) with $x = c$: $f(c^2) = c + f(c)^2 = c$, consistent. Now compute $f(f(0)) = 0 + d^2 = d^2$ from Step 1; and substituting $x = 0, y = 0$ into the original gives $f(f(0)) = 0 + f(0)^2 = d^2$ as well — consistent, so we use a different lever. From (\dagger), $f(x^2) \geq c$ for all $x$ is false in general since $f(x)^2 \geq 0$ only gives $f(x^2) \geq c$; thus $f(t) \geq c$ for every $t \geq 0$. Apply $f$ to $f(c) = 0$: by Step 1, $f(0) = f(f(c)) = c + d^2$, i.e. $d = c + d^2$ again. To finish, note from (\ast) the map $y \mapsto c^2 + f(y)$ is the inverse of $f$; since $f$ is also its own "shifted" inverse via $f(f(y)) = y + d^2$, comparing gives, for all $y$, $c^2 + f(y) = f^{-1}(y)$ and $f^{-1}(y) = f(y) - d^2$ (the latter by applying $f^{-1}$ to $f(f(y)) = y + d^2$). Hence $c^2 + f(y) = f(y) - d^2$, giving $c^2 = -d^2$, which forces $c = d = 0$. Step 4 (additivity and finish). With $c = 0$: (\dagger) becomes $f(x^2) = f(x)^2$ (so $f \geq 0$ on $[0, \infty)$), $f(0) = 0$, and $f(f(y)) = y$ (an involution). The original is $f(x^2 + f(y)) = y + f(x)^2 = y + f(x^2)$. Replace $y$ by $f(y)$ (legal since $f$ is onto) and use $f(f(y)) = y$: $f(x^2 + y) = f(y) + f(x^2)$. As $x^2$ ranges over $[0, \infty)$, $f(t + y) = f(t) + f(y)$ for all $t \geq 0$, which extends to full additivity. An additive $f$ that is nonnegative on $[0, \infty)$ is monotone, hence linear: $f(x) = kx$. Then $f(f(y)) = k^2 y = y$ gives $k^2 = 1$, and $f \geq 0$ on $[0, \infty)$ rules out $k = -1$, leaving $k = 1$. Therefore $f(x) = x$, which satisfies $f(x^2 + f(y)) = x^2 + y = y + f(x)^2$.

**Second method.** Streamlined finish after $c = 0$: from $f(x^2 + f(y)) = y + f(x)^2$ and $f(x^2) = f(x)^2$, put $x = 0$ to get the involution $f(f(y)) = y$. Replacing $y$ by $f(y)$ yields $f(x^2 + y) = f(x^2) + f(y)$; additivity plus $f \geq 0$ on $[0, \infty)$ (since $f(t) = f(\sqrt{t})^2 \geq 0$ for $t \geq 0$) gives monotonicity, hence $f(x) = x$. The whole problem hinges on extracting bijectivity first — that is what licenses choosing $c$ and substituting $f(y)$ for $y$.